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Randomized prospective study of ACL reconstruction with interference screw fixation in patellar tendon autografts versus femoral metal plate suspension and tibial post fixation in hamstring tendon autografts: 5-year clinical and radiological follow-up results

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Abstract Patellar tendon graft has been the most frequently used material in anterior cruciate ligament (ACL) reconstruction, but the hamstring tendons have been increasingly used as well; however, which graft is to be preferred is not adequately supported by existing clinical studies. In this prospective randomized clinical trial, the study hypothesis was that the hamstring tendons are equally good graft material as the patellar tendon in ACL reconstruction. Ninety-nine patients with laxity due to a torn ACL underwent arthroscopically assisted reconstruction with graft randomization according to their birth year to either patellar tendon with metal interference screw fixation or double looped semitendinosus and gracilis tendons with fixation similar to the Endobutton technique using a titanium metal plate suspension proximally and screw-washer postdistally. Excluding preoperative Lysholm knee score, there were no significant differences between the two groups in the preoperative and operative data. A standard rehabilitation regimen was used for all the patients, including immediate postoperative mobiliza-

tion without a knee brace, protected weight bearing for 2 weeks, and return to full activity at 6–12 months postoperatively. Forty patients in the patellar tendon group and 39 patients in the hamstring tendon group were available for clinical evaluation at median 5 years after surgery (ranges 3 years 11 months–6 years 7 months). The results revealed no statistically significant differences with respect to clinical and instrumented laxity testing, isokinetic muscle torque measurements, International Knee Documentation Committee ratings, Lysholm (knee score), Tegner (activity level) and Kujala patellofemoral knee scores. There was an enlargement of the drill tunnels, statistically more in the hamstring tendon group, but no increase from 2 to 5 years in either group. Narrowing of the joint spaces (IKDC measurement method) from 2 to 5 years postoperatively was seen in both the groups, however, without difference between the two groups.

Keywords Anterior cruciate ligament · Patellar tendon · Hamstring tendon · Randomized study

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Introduction

Good results after reconstruction with autogenous tendon grafts have been reported in symptomatic

anterior cruciate ligament (ACL) insufficiency. The two most commonly used autogenous grafts for intra-articular ACL reconstruction are the central third of the patellar tendon (BPTB) and the semitendinosus

and gracilis (STG) tendons harvested from the hamstrings.

Postoperative disadvantages of the BPTB graft include patellofemoral pain or donor site discomfort, quadriceps muscle weakness, and possible rupture of the patellar tendon and patellar fracture [1, 31, 40]. Proposed disadvantages among the STG grafts include failure to achieve immediate rigid fixation to bone and lower stiffness compared with a BPTB graft or the native ACL [28, 35, 41, 44]. According to meta-analyses of studies of ACL reconstructions, there is a tendency to increased laxity in the STG reconstructions [20, 46]. It has also been reported that hamstring weakness at greater flexion angles of the knee can be a problem after STG reconstruction [34]. Radiographic tunnel widening has also been a concern in the STG reconstructions, although it seems not to affect the postoperative stability [15, 26, 43]. Long-term effects of the tunnel expansion are unknown, but it might complicate revision surgery [19].

There are numerous reports of ACL reconstruction in the literature with increasing number of randomized prospective clinical trials recently published comparing patellar and hamstring tendon grafts [4, 6, 8, 9, 13, 17, 27, 31, 42].

The purpose of this study was to report the clinical outcome, instrumented laxity and isokinetic muscle output measurements, subjective evaluation in terms of knee scores and radiological results after 5 years follow-up in two arthroscopically assisted ACL reconstruction techniques: patellar tendon (BPTB) and hamstring tendon (STG) autografts. Results after 2 years follow-up have been reported earlier [27].

Materials and methods

Patients and entry criteria

Ninety-nine patients were enrolled in this prospective study of primary arthroscopically assisted ACL reconstruction and graft randomization into a BPTB or STG group according to their birth year (even year = patellar tendon, odd year = hamstring tendons). The time interval from injury to operation was 19 months (ranges, 1 week–20 years) in the patellar tendon group ($n=51$) and 16 months (ranges 2 weeks–10 years) in the hamstring tendon group ($n=48$, *NS*).

Reconstruction of the torn ACL was indicated when diagnosed on clinical examination and confirmed at arthroscopy in an otherwise healthy patient who experienced instability in physical activities or wished to maintain his or her preinjury level of activity. Patients with all kinds of meniscal tears and chondral lesions were allowed to enter the study, as well as those with collateral ligament tears. Revision surgery and

concomitant posterior cruciate ligament tears were the exclusion criteria. Of the original 105 patients randomized into the study, six patients were eventually excluded as they had a torn ACL in their contralateral knee as well.

The study was approved by the local ethical committee and the patients gave their informed consent to participate.

Surgical technique and postoperative rehabilitation program

Two senior authors (JS and AH) performed all the reconstructions using both techniques equally. The patients were examined under anesthesia (Lachman, drawer, pivot shift, and rotation tests), after which arthroscopy confirmed the diagnosis of ACL tear and necessary meniscal surgery, either partial resection or repair was performed, followed by the ACL reconstruction with a technique determined by the birth year. Direct repair was done in the case of medial collateral tear.

In the BPTB group, the surgical procedure was an “outside-in” arthroscopically assisted two-incision technique with the central third of the patellar tendon used as a free autograft with matching drill tunnels made in the femur and tibia [11] (Fig. 2a). Specially designed “rear-entry” drill guides were used on the femoral side (Linvatec, Largo, FL, USA). The femoral entry point was targeted to 11 O’clock in the right knee and to 1 O’clock in the left knee. The fixation of the graft in the drill tunnels (9 or 10 mm) was performed with interference metal screws of 8 or 9 mm (Linvatec).

In the STG group, an arthroscopic one-incision transtibial technique with double looped semitendinosus and gracilis tendons was used. The drill tunnels of 8 or 9 mm in diameter were done using drill guides to confirm the correct position of the tunnels. To find the femoral entry point, we used the “bull’s eye” drill guide (Linvatec) aimed to similar position as in the BPTB group. Proximally, a suspension type graft fixation similar to the Endobutton method [7, 39] was done with a small metal plate cut from an AO titanium plate (AO, Bern, Switzerland, Fig. 3a). Dacron loop (Davis & Geck, Danbury, Connecticut) was used to connect the hamstring tendon graft to the plate. The tensile strength of the Dacron loop was 11.3 kPa according to the manufacturer. Distal fixation was achieved by tying the graft around a cortical 4.5 mm screw secured with a spiked washer over the graft ends (AO).

Postoperatively, no knee braces were used unless medial collateral ligament (MCL) surgery was done or a torn MCL was treated with a brace. The rehabilitation protocol was identical in both the groups and all the knees were immediately mobilized with full weight

bearing allowed 2 weeks postoperatively. A closed kinetic chain muscle rehabilitation program was adopted. Active quadriceps muscle activity was delayed until 3–4 weeks postoperatively. Cycling and aqua jogging were started 4–6 weeks postoperatively and jogging was allowed 6 weeks postoperatively. About 6 weeks postoperatively a 1-day rehabilitation visit was arranged at the physiotherapy department. Clinical follow-up was done at 3 weeks, 6 weeks, 3 months, and 6 months postoperatively to monitor the patients' motivation for the rehabilitation. The patients were allowed to return to sports gradually, usually without limitations 6–12 months postoperatively.

Clinical assessments and follow-up evaluations

Clinical evaluations of knee function and laxity were performed by the two senior authors (JS and AH) preoperatively and at 1, 2, and 5 years postoperatively. The side-to-side laxity difference on manual examination was graded into three categories where - means 0–3 mm difference (normal), + means 3–5 mm difference (slight), and ++ means over 5 mm difference (clear). The pivot shift test was graded - as normal, + a glide, and ++ a clear pivot shift.

To determine objective anteroposterior knee stability, physiotherapists used the CA 4000 arthrometer (OSI, Hayward, CA, USA) and a dynamometer (Lido Multi-joint II, Loredan Biomedical Inc., West Sacramento, CA, USA) to measure isokinetic muscle torque (0–80° of knee flexion) of the flexor–extensor system of both the lower extremities. Patients completed the Lysholm knee score, Tegner activity level [45], and Kujala patellofemoral [30] evaluation forms at 1, 2, and 5 years

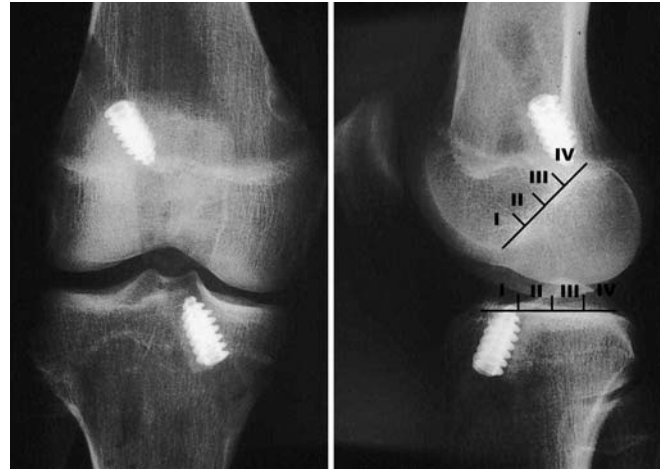


Fig. 1 X-ray of a knee with ACL reconstruction, division into zones 1–4 along the Blumensaat line in the femur and in the proximal tibia is shown in the side view on the right

postoperatively and the International Knee Documentation Committee (IKDC) [22] classification was done as well. From the postoperative supine radiographs, the location of the drill tunnels was evaluated by dividing the Blumensaat line in the femur and proximal end of the tibia into four zones (1 through 4), with zone 1 being the most anterior (Fig. 1). The ideal position would be zone 4 in the femur and zone 2 in the tibia [21]. Anteroposterior and lateral weight bearing radiographs in extension of the operated knee were obtained 2 and 5 years postoperatively. The radiographic evaluation of the joint space was done according to the IKDC recommendations. In grade 1 the joint space is >4 mm, in grade 2 the joint space is also >4 mm but there are

Fig. 2 a Anteroposterior radiograph taken 1 day postoperatively showing the patellar tendon ACL reconstruction technique. **b** The same case as in Fig. 2a 5 years postoperatively with obvious medial knee joint degeneration

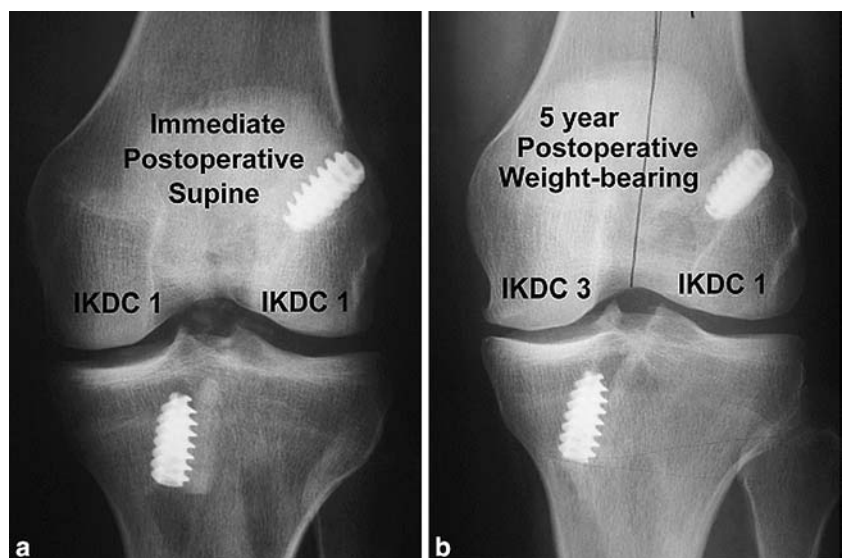
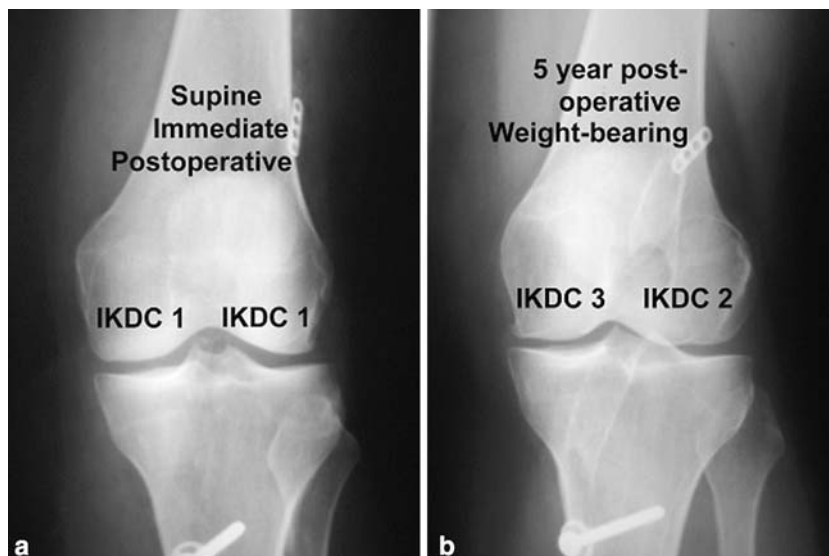


Fig. 3 **a** Anteroposterior radiograph taken 1 day postoperatively showing the hamstring tendon ACL reconstruction technique. **b** The same case as in Fig. 3a 5 years postoperatively with obvious medial knee joint degeneration



osteophytes or sclerosis as well, in grade 3 the joint space is 2–4 mm, and finally in grade 4 the joint space is <2 mm. The radiographic measurements were done by an author not involved in the treatment of the patients (EL).

Statistical methods

Statistical analysis was performed with the BMDP statistical package (Statistical Solutions Ltd., Cork, Ireland). Differences between the groups were tested with the chi-square test, Student's *t*-test, and analysis of variance. Pair wise *t*-test and Wilcoxon tests were used to test repeated measurements within the groups over time. Correlations were tested with a regression analysis. The minimum level of significance was $P=0.05$. The Lysholm score was one of our main outcome measurements. With a difference of ten points, a clinically significant difference between the treatment groups would be acknowledged. Thus, by accepting less than a 5% probability of a type I error and a power of 80%, a sufficient total sample size would be 54. Having collected a total of 99 cases, we believe that the statistical power was adequate.

Results

Preoperative evaluation

There were no statistically significant differences between the preoperative data of the two groups with respect to gender, age, time from injury to operation, knee laxity tests, Tegner activity levels, Kujala patellofemoral

scores, or isokinetic muscle torque values. In the BPTB group, the mean Lysholm score was slightly higher than in the STG group (74 vs. 68, $P=0.044$). The female patients were slightly older (mean 33 years) than the males (mean 29 years, $P=0.0454$).

Associated injuries and treatment

There were nine MCL tears, of which five patients had grade 2 and four patients had grade 3 MCL tears.

Five of these patients were in the BPTB group and four in the STG group. Two grade 3 MCL tears were repaired at the time of the ACL reconstruction (one patient from each group). The rest of the patients with MCL tears were given a knee brace to be worn preoperatively for 6 weeks if the reconstruction was not performed in the acute phase. All the MCL tears healed and the knees were stable to valgus stress on clinical examination throughout the follow-up period.

At the time of the ACL reconstruction, 28 meniscal repairs or resections were performed in 26 patients. In the BPTB group, there were three partial resections and five repairs of the medial meniscus and six resections and two repairs of the lateral meniscus. In the STG group, there were four resections and four repairs medially and three resections and one repair laterally. There were fresh osteochondral lesions in eight patients (two in the BPTB group and six in the STG group). The lesions were from 5×5 to 10×10 mm in seven patients and they were treated with the microfracture technique. In one patient (hamstring tendon group), the lesion was 30×40 mm in size and it was treated with the autologous chondral cell culture technique immediately after the ACL reconstruction (during the same operation). Sec-

and look arthroscopy performed 1 year later disclosed nearly complete coverage of the lesion.

Complications and additional surgery during the five-year follow-up

In the BPTB group, there was one case with clinical presentation as a purulent arthritis 1 week after the ACL reconstruction with the bacterial culture being, however, negative. Arthroscopic debridement and profuse lavage was performed with uneventful recovery. The knee remained stable throughout the follow up. There were no deep vein thromboses in either group. Scar revisions at the harvest site were done in three cases. In the STG group, the tibial AO fixation post was removed by 5 years postoperatively in 34 cases because the patients had discomfort at the fixation site. The interference screw in the BPTB group was removed from the tibia in two cases and from the femur in one case. Omitting the AO screw removals, there was no statistical significant difference in the number of postoperative procedures between the groups or with respect to individual findings. Additional surgery was not done during the 5-year follow-up period in 50 cases. One procedure (screw removal and/or arthroscopy) was done in 27 cases, two procedures in 12 and three in 10 cases. A scar formation adjacent to the graft ("cyclops") was removed 13 times (11 times in eight BPTB cases and two times after STG reconstruction, NS). Other arthroscopic scar removals were done in three cases (all in BPTB group). Various chondral operations (microfracture, osteochondral allograft transplantation, revision, and loose body removal) were done ten times in seven cases (two BPTB and five STG cases). Meniscal tear was found in 11 arthroscopies done on nine patients during the 5-year follow-up time (three patients in BPTB and six in STG group). Two medial meniscus tears were sutured (one in each group) and seven resected (three in BPTB and four in STG group). Three lateral meniscus tears were treated with resection (all in STG group). Two manipulations were performed in BPTB under anesthesia and one in the STG group and in one case a high tibial osteotomy to correct weight bearing alignment was done (STG case). There were no specific findings in three arthroscopies (two patients in the STG group).

Five-year follow-up

Of the 99 patients in the study, 79 (80%) were available for the median 5-year follow-up (ranges 3 years 11 months–6 years 7 months), 40 patients in the BPTB group and 39 in the STG group. Sixteen of the 20 missing cases could not be contacted or they were not willing to attend mainly because of long traveling dis-

tances. Four revisions were done, two because of a new high-energy trauma, which had disrupted the grafts (both in the STG group). Two knees of the STG group became loose without any obvious reason and they can be classified as graft failures. All four cases were revised during the 5-year follow-up period and their data after the revision procedure were excluded because the result would not represent that of the original reconstruction. There was an ACL injury in the contralateral knee in five patients (one in the BPTB and four in the STG group) and an ACL reconstruction was done during the 5-year follow-up period in all four patients in the STG group. After the contralateral injury, the instrumented laxity measurements of the control knee done at the subsequent follow-up visits were omitted from the analysis.

Clinical data and instrumented testing

At the 5-year follow-up examination, the mean range of motion was -4.8° to 147.4° in the BPTB group and -5.8° to 146.1° in the STG group (NS). On clinical examination at the 5-year follow-up, the Lachman test was one + or two + in eight patients in the BPTB group and in six patients in the STG group (Table 1). There was one + or two + positive pivot shift test in four patients in both the groups (Table 2). Only one patient (from the STG group) was clearly unstable in the Lachman and pivot shift tests, the rest were one +.

The mean side-to-side difference in anteroposterior knee laxity was 1.5 mm in the BPTB group and 1.2 mm in the STG group (NS, Table 3). There was no statistical difference between the 2- and 5-year results. There were no statistically significant differences in the clinical data

Table 1 Lachman test 5 years postoperatively after ACL reconstruction with patellar (BPTB) and hamstring tendons (STG), Chi-square test

	No. (–)	Slight (+)	Clear (++)	Total
BPTB	29	8	0	37
STG	31	5	1	37
Total	60	13	1	74

$$\chi^2 = 0.41$$

Table 2 Pivot shift test 5 years postoperatively after ACL reconstruction with patellar (BPTB) and hamstring tendons (STG), Chi-square test

	No. (–)	Slight (+)	Clear (++)	Total
BPTB	33	4	0	37
STG	33	3	1	37
Total	66	7	1	74

$$\chi^2 = 0.56$$

Table 3 Side-to-side laxity difference (CA-4000) and isokinetic muscle output (Lido, peak torque, percentage from the control knee) 5 years postoperatively after ACL reconstruction with patellar (BPTB) and hamstring tendons (STG)

	BPTB	STG	Significance, <i>t</i> -test
Side-to-side laxity difference, mm $\pm$ SD	1.5 $\pm$ 3.0	1.2 $\pm$ 3.9	<i>P</i> = 0.72
Extension torque 60°/s, %	88.2 $\pm$ 16.0	89.2 $\pm$ 13.3	<i>P</i> = 0.78
Extension torque 180°/s, %	93.6 $\pm$ 19.7	91.4 $\pm$ 11.0	<i>P</i> = 0.56
Flexion torque 60°/s, %	96.8 $\pm$ 14.6	97.5 $\pm$ 12.1	<i>P</i> = 0.84
Flexion torque 180°/s, %	95.5 $\pm$ 11.3	96.5 $\pm$ 12.7	<i>P</i> = 0.69

and instrumented testing results between women and men.

The mean time from injury to operation was 4.3 years in the eight patients with a positive pivot shift test (+, *n* = 7; ++, *n* = 1) 5 years postoperatively, whereas those patients with a negative pivot shift test underwent surgery at a mean of 1.3 years after injury (*P* = 0.0276).

Muscle strength

At the 5-year follow-up, there were no statistically significant differences between the groups with respect to isokinetic peak muscle torque when the percentages injured/control were analyzed (Table 3). Extension torque at 60 and 180°/s remained 11 and 6–8% reduced when compared to the control side, however, without statistical difference between the groups. Flexion torque was reduced only by 2–4%. There were no statistically significant differences in the muscle strength percentages between women and men.

The extension torque percentage at 60 and 180°/s diminished from IKDC A–IKDC C categories (Table 4). Extension and flexion torque percentages at the 5-year follow-up (injured/control at (60 and 180°/s) did not have a statistically significant correlation with the Lysholm knee score or side-to-side laxity difference, but there was a statistically significant (*P* < 0.05) positive correlation with the Tegner activity level and the Kujala patellofemoral score in the extension torque at 60 and 180°/s and in the flexion torque at 180°/s. Flexion torque at 60°/s correlated negatively (without statistical significance) with all of the above-mentioned variables. The same phenomenon was also seen with regard to the IKDC score, i.e. in the worst IKDC classification (C), the best flexion torque was seen at 60°/s (Table 4).

Table 4 Isokinetic peak muscle torque percentages from the control knee 5 years postoperatively in the patellar and hamstring tendon groups broken down into the IKDC categories

C	Extension 60°/s	Extension 180°/s	Flexion 60°/s	Flexion 180°/s
A	91.4%	93.5%	94.8%	95.1%
B	89.2%	94.4%	97.6%	96.7%
C	82.4%	84.7%	99.5%	95.3%
Significance	0.2258	0.1752	0.6082	0.8651

IKDC scores

At the 5-year follow-up, there were no statistically significant differences between the groups with respect to overall IKDC scores, the majority belonging to A and B categories (Table 5). There was no statistical difference between the 2- and 5-year results.

Lysholm knee score

At the 5-year follow-up, there were no statistically significant differences between the groups with respect to the Lysholm knee scores (Table 6). The median Lysholm knee score was 95 (“excellent”) in both the groups. There were no statistically significant differences in the Lysholm score between females and males or comparing the 2-year results with the 5-year results.

Kujala patellofemoral score

At the 5-year follow-up, there were no statistically significant differences between the groups with respect to the Kujala patellofemoral score (maximum 100) with median 96.5 in the patellar tendon and 98 in the hamstring tendon groups, respectively (Table 6). There were no statistically significant differences in the Kujala patellofemoral scores between females and males or between the 2- and 5-year results.

Tegner activity level

The 5-year postoperative median Tegner activity level was six (ranges 1–10) in the BPTB group and six (ranges

Table 5 IKDC classification 5 years postoperatively after ACL reconstruction with patellar (BPTB) and hamstring tendons (STG), Chi-square test

	A	B	C	Total
BPTB	13	19	6	38
STG	9	22	6	37
Total	22	41	12	75

$$\chi^2 = 0.62$$

Table 6 Knee scores, median, and range 5 years postoperatively after ACL reconstruction with patellar (BPTB) and hamstring tendons (STG)

	BPTB	STG	Significance, <i>t</i> -test
Tegner	6, 1–10	6, 3–9	<i>P</i> =0.57
Lysholm	95, 34–100	95, 34–100	<i>P</i> =0.52
Kujala PF	96.5, 47–100	98, 54–100	<i>P</i> =0.58

3–9) in the STG group (Table 6). Level 6 makes it possible to play various ball games and do downhill skiing. The patients in either group did not reach their preinjury median Tegner activity level of seven (*P*<0.001). There were no statistically significant differences between the groups or between the 2- and 5-year results with respect to the Tegner activity level. Women had a lower Tegner activity level than men before the injury (*P*=0.0007) and it remained lower throughout the follow-up, however, at 5 years it was only marginally lower (*P*=0.057).

Radiographic evaluation

The postoperative radiographs (*n*=96) showed that the drill tunnels were placed in the anterior part of the tibia more often in the patellar tendon group than in the hamstring tendon group. In the patellar tendon group, 14 of 50 tibial tunnels were in zone 1, whereas in the hamstring tendon group 3 of 46 were located in zone 1 (*P*=0.0059). No statistical difference was noted with respect to the femoral tunnel placement where 11 out of 50 tunnels in the patellar tendon group were in zone 3. The corresponding numbers for the hamstring group was four out of 46 (*P*=0.0729). There were no differences in the knee scores or side-to-side laxity with respect to the placement of drill tunnels.

At 2- and 5-year radiographic examination drill tunnel enlargement was greater in the hamstring group. The difference was statistically significant at 2 years on the femoral side and at 5 years in the tibia (Table 7). There was a slight decrease in the drill tunnel widths in both the patellar tendon and hamstring groups from 2- to 5-year follow-up, but this was not statistically significant. The tunnel enlargement did not correlate in any

Table 8 Knee joint space measured according to IKDC from 2 to 5 years postoperatively after ACL reconstruction with patellar (BPTB) and hamstring tendons (STG). Weight bearing radiographs, mean and standard deviation (Wilcoxon test)

	BPTB and STG	Significance
Medial tibiofemoral	1.4±0.5 → 1.9±0.8	<i>P</i> <0.0001
Lateral tibiofemoral	1.1±0.3 → 1.3±0.5	<i>P</i> =0.0005
Patellofemoral	1.3±0.5 → 1.6±0.5	<i>P</i> <0.0001

way to the final result with respect to laxity or knee scoring.

Weight bearing radiography was done at 2 and 5 years postoperatively. Preoperative and immediate postoperative radiographs were usually supine, which makes them unsuitable to compare to those taken 5 years postoperatively. IKDC measuring method of knee joint space disclosed no difference between the BPTB and STG groups preoperatively or 2 and 5 years postoperatively. When the 2 and 5 years postoperative radiographs were compared, a statistically significant worsening of the radiological status was seen in all compartments of the knee joint (Table 8, Figs. 2b,3b).

In the 5 years postoperative radiographs, the medial compartment joint space status was worse in the knees with medial meniscus pathology (torn or previously resected, *n*=17) at the time of the ACL reconstruction (resected or repaired) when compared to the knees without meniscus pathology (*n*=54, *P*=0.0226). Lateral meniscus pathology did not have the same effect (Table 9).

Medial joint space degeneration at 5 years postoperatively correlated with age and with the length of time from injury to reconstruction, but not with instrumented laxity measurements at 5 years. Patients were divided into two equally large groups with respect to their age at the time of reconstruction, the cut point being 30.38 years. At the five-year follow-up, younger patients had less medial joint space narrowing than older patients. When looking at the change in the knee joint space according to the IKDC measuring method from 2 to 5 years in the patients below the cut point, the degeneration had advanced by 0.2 IKDC units and in those above the cut point, the advancement was by 0.7 units (*P*=0.0026). The degeneration was thus more rapid in the older age group.

Table 7 Drill tunnel width and standard deviation measured 2 and 5 years postoperatively after ACL reconstruction with patellar (BPTB) and hamstring tendons (STG)

	Drill tunnel width 2 years postoperatively			Drill tunnel width 5 years postoperatively		
	BPTB	STG	Significance, <i>t</i> -test	BPTB	STG	Significance, <i>t</i> -test
AP femoral	11.3±2.3	13.3±1.9	0.0002	10.9±2.3	13.1±1.6	<i>P</i> =0.1025
AP tibial	11.5±2.4	12.0±1.8	0.3650	11.0±2.2	12.3±2.1	<i>P</i> =0.0180
Sagittal tibial	11.0±2.1	11.7±1.6	0.1024	10.4±2.7	11.8±1.8	<i>P</i> =0.0138

Table 9 Meniscus status, intact or pathology (torn or previously resected) at the time of the ACL reconstruction and weight bearing joint space measured according to IKDC 5 years postoperatively after ACL reconstruction with patellar and hamstring tendons, mean and standard deviation (*t*-test)

	Joint space according to IKDC		Significance
	Meniscus intact	Meniscus pathology	
Medial tibiofemoral	<i>n</i> = 54, 1.7 ± 0.8	<i>n</i> = 17, 2.2 ± 0.8	<i>P</i> = 0.0226
Lateral tibiofemoral	<i>n</i> = 60, 1.3 ± 0.5	<i>n</i> = 11, 1.3 ± 0.6	<i>P</i> = 0.9527

There were no statistically significant differences in the IKDC knee joint space measurements between women and men in any of the three knee compartments.

Discussion

Hamstring tendon ACL reconstruction with Endobutton or metal plate fixation has been reported [7, 24, 32, 38, 39] and compared with patellar tendon graft in six studies [6, 14, 17, 25–27].

Enlargement of the drill tunnels have been seen in both the patellar tendon and especially in hamstring tendon and allograft reconstructions [10, 15, 25, 26, 32], but the long-term effects are still unknown [19, 43]. We have reported the radiological phenomenon of bone tunnel enlargement in patients with hamstring tendon autograft, which did not correlate with postoperative instability [26]. L'Insalata et al. [25] did not compare the clinical outcome between the patellar and hamstring tendon reconstruction, but in follow-up radiographs there was enlargement of the bone tunnels in the hamstring group after the Endobutton fixation method. It has been proposed that fixation close to the joint would diminish the tunnel widening, but contrary to this tunnel enlargement was greater in hamstring grafts with interference screw anatomic fixation than in extracortical fixation [10, 32]. Our finding of decrease in the tunnel widening with time is also supported by others [10, 43].

Several clinical follow-up reports of randomized studies comparing patellar and hamstring tendon reconstruction have been published [2–4, 6, 8, 9, 13, 14, 17, 31, 33, 36, 42].

Marder et al. [33] reported that the only difference between the patellar and hamstring tendon groups (reconstructed on one-to-one alternating basis) was a weaker flexion torque in the latter group. Both graft types were fixed by sutures around the femoral and tibial post and washer.

Aglietti et al. [2] published a material with 63 ACL reconstructions, alternating randomization done with patellar or four strand hamstring tendon grafts. In the

patellar tendon group, they used outside-in press fit fixation on the femoral side and tied four non-absorbable sutures over a cortical screw and washer. The tibial side was fixed with an interference screw and four non-absorbable sutures tied over a cortical screw and washer. In the hamstring tendon group, cortical screw post fixation was used. After mean 5.7 years (ranges 5.2–6.2 years) 60 patients were seen. They found out that there was better stability in the patellar tendon group and therefore they recommended patellar tendon grafts for patients with sports activity. Hamstring graft would suit elderly patients and those with less physical activity when they already have symptoms of anterior knee pain and it would of course suit as a graft material in revisions after patellar tendon reconstruction.

O'Neill [36] published a material of 225 patients with a mean follow-up of 102 months (ranges 6–11 years). He had three treatment groups (randomized according to the month of birth): arthroscopically assisted two-incision patellar or hamstring tendon graft and single-incision patellar tendon graft. He used doubled rather than quadrupled hamstring tendon grafts and barbed staples were used to fix the autograft. The operative technique in the two-incision patellar tendon group was similar to that in our study. In the follow-up, there were no differences between the groups with regard to the rate of graft failure, the amount of knee laxity or the functional outcome. About 11.6% of the knees had radiographic evidence of degenerative arthritis without, however, difference with regard to the choice of graft or the technique of reconstruction.

Beynon et al. [9] used metal interference screw fixations in patellar tendon grafts and staple fixation in two-strand hamstring grafts. After 3-years follow-up, the results in patellar tendon group were better in terms of laxity and flexion torque, but the functional results did not differ.

Shaieb et al. [42] randomized their patellar and quadrupled hamstring tendon graft patients according to the birth date. Fixation with metal interference screw was similar in both the groups. About 2-years follow-up results were similar in terms of knee scores and laxity, but there were more patellofemoral pain and loss of motion in the patellar tendon group. They also reported two failures in each group, of which three were revised. Similar design was used by Beard et al. [8], but their follow-up lasted only 1 year postoperatively in 75%, with no significant difference between the groups.

Ejerhed et al. [13] used metal interference screws in a prospective randomized (sealed envelope) study in both the patellar tendon and quadrupled semitendinosus grafts. The only significant difference after the 2 years follow-up was an impaired knee walking ability in the patellar tendon group.

Aglietti et al. [3] reported a randomized study (alternation manner) of patellar tendon with proximal

fixation using a transcondylar Tunneloc (Arthrotek) screw and distally with a metal interference screw compared with four bundle hamstring graft fixed with a proximal transcondylar Bone Mulch screw and distally with a WasherLoc (Arthrotek) post fixation. At the 2-year follow-up evaluation, no differences were found in terms of the knee scores, laxity, or muscle strength. In the patellar tendon group, they found a higher prevalence of postoperative kneeling discomfort and an increased area of decreased skin sensitivity. In the hamstring tendon group, they recorded a higher prevalence of femoral tunnel widening. In this group, a correlation was also found between medial meniscectomy and an increased prevalence of the pivot-shift glide. With the use of accurate and proven surgical and rehabilitation techniques, they believe that both the grafts are an equivalent option for ACL reconstruction.

Laxdahl et al. [31] compared prospectively in a controlled randomized study patellar tendon and 3- or 4-strand hamstring tendon ACL reconstructions. The 2-years follow-up results were similar with respect to laxity (*KT* 1000) and knee scores (Tegner, IKDC, and Lysholm). There was significantly more discomfort at the donor site in the patellar tendon group 2 years postoperatively when compared to the hamstring tendon groups and compared to the preoperative situation within the patellar tendon group.

In addition to the above-mentioned randomized studies, three prospective materials have been published describing patellar tendon and hamstring tendon grafts in ACL reconstruction [12, 16, 37]. Corry et al. [12] reported more laxity in women and less pain on kneeling in the hamstring grafts. Feagin et al. [16] and Otero and Hutcheson [37] concluded, based on their results, that both grafts are useful but the patellar tendon grafts provide better stability.

There are also randomized prospective studies of the suspension type (Endobutton or metal plate) hamstring tendon fixation [6, 14, 17, 27]. Aune et al. [6] randomized their ACL reconstructions to either patellar tendon ($n=35$, metal interference screw fixation) or four-strand hamstring tendon ($n=37$, proximally Endobutton and distally metal interference screw fixation) grafts. The 2-years follow-up disclosed no differences in terms of laxity or knee scores. In the hamstring group, there was a weaker flexion torque and less anterior knee pain ("kneeling pain").

In a randomized, prospective study by Eriksson et al. [14] there were 164 patients, in whom the operative technique in the patellar tendon group was identical to ours, whereas their hamstring tendon graft was a 4-fold semitendinosus graft, which was proximally fixed with the Endobutton plate and distally using a double post fixation. After a mean 31 months follow-up (ranges 24–59 months) they concluded that with respect to all

variables describing the end result, there was no difference between the groups.

Feller and Webster [17] did a prospective (random number randomization) study on patellar and hamstring tendon reconstructions. Both groups received an Endobutton fixation on the femoral side and distally they used metal interference screw in the patellar tendon group and a post fixation in the hamstrings. The 3 years follow-up results were satisfactory according to the functional outcome in the both groups, but with increased morbidity in the patellar tendon group and increased knee laxity and radiographic femoral tunnel widening in the hamstring tendon group.

Recent meta-analyses of studies published until 2,000 have shown that patellar tendon reconstructions yield better correction of abnormal laxity compared to hamstring tendon grafts, but there are more complications and postoperative anterior knee pain problems [20, 46]. The percentages for graft failures were given 1.9% for the patellar tendon graft and 4.9% for hamstring tendon grafts [20] ($P<0.001$), the weak point of their conclusion being missing the definition of graft failure.

In our study, the patellar tendon and hamstring grafts were in fact not compared with each other because the fixation methods differed. The same type of fixation, metal interference screws on both the sides (proximally "inside-out" inserted), was used in three studies comparing patellar and hamstring tendon autograft reconstructions [12, 13, 42] and there were no major differences in the results between the groups.

We have used a "two-incision" technique in patellar tendon reconstructions so as not to cause intra-articular aperture impingement between the graft and the screw, which obviously may occur when endoscopic (in-out) screw fixation is used. The other technique we have used, metal plate fixation, in the hamstring tendon grafts is developed from inside the joint to be inserted and there is no risk for similar "impingement" problem. The tibial fixation was done with an AO screw and an AO spiked washer, which being so bulky had to be removed in 34 cases. Other types of post fixation might not have to be removed [38].

The mean time from injury to reconstruction was statistically significantly longer in the eight patients with a positive pivot shift (+ or ++) test 5 years postoperatively than in those patients with a negative pivot shift. Therefore, it seems that stabilization of the ACL-deficient knee should be done without unnecessary delay, a conclusion also supported by Eriksson et al. [14].

Contrary to Corry et al. [12], we did not see worse results in the female patients in terms of clinical or instrumented stability tests. The knee scores were also comparable. The Tegner activity levels were lower, but they were lower also before the injury.

The data on the isokinetic peak muscle torque measurements are difficult to interpret. It is rather surprising

that there were no differences between the study groups with respect to the graft source, with the hamstring tendon group also lacking 11% extension peak torque 5 years postoperatively. The positive correlations with Tegner activity level and Kujala patellofemoral score are logical. The measurements in our study were done through 0–80° of knee flexion and the peak torque was picked up from the curve at variable flexion angle. Nakamura et al. [34] have presented a study of hamstring tendon ACL reconstruction where at 90° of flexion the knee flexor strength was less effectively restored as compared to a flexion angle where the peak torque was generated.

In the weight bearing radiographs obtained at the 5 years follow-up visit, the narrowing of the joint space was noted causing suspicion to the development of arthrosis. IKDC measuring method is probably only suggestive, because the joint space is roughly divided into four categories depending on the cartilage thickness between the femur and tibia. There is a possibility for errors depending on the position of the joint between the examinations 2 and 5 years postoperatively. The pre-operative and immediate postoperative radiographs were not always weight bearing, but the 2- and 5-year radiographs were taken with identical imaging technique. On the other hand, the IKDC classification takes also into account the osteophytes and sclerosis, which can be seen also on non-weight bearing radiographs. There was no difference in the development of the knee joint space narrowing between the groups.

Five and seven years follow-up studies [18, 29] have shown that arthritic changes will develop even with intact menisci at the time of ACL reconstruction [29], but is more likely if the menisci were damaged. O'Neill [36] noted radiographic evidence of arthrosis in 11.6% at mean 6 years after ACL reconstruction, but no significant relationship with meniscal surgery was seen. In a recently published 10-year follow-up study of press fit

BPTB reconstruction, joint space narrowing was seen in 15–25% in various knee compartments [23]. Asano et al. [5] did a second look arthroscopy after double bundle Endobutton ACL reconstruction at the time of removal of the tibial post fixation and found that there are signs of degeneration in all the articular surfaces except the lateral femoral condyle as early as 15 months postoperatively. The risk factors were female gender and age over 30 years. Meniscal tears at the time of the reconstruction were associated with joint surface damage; however, no worsening was seen in these cases at the second look arthroscopy. Our results showed a correlation between the age of the patients and knee joint degeneration because the worsening in IKDC joint space grading between 2 and 5 years was greater in the older age group. Also the length of time from injury to reconstruction correlated with degenerative changes. Therefore, it might be reasonable not to delay the reconstruction unnecessarily.

The results of the present study and of others do not confirm the superiority of either the patellar or hamstring tendon grafts in ACL reconstruction. According to the literature, there is a tendency toward a little more laxity in the hamstring reconstructions, but the increased morbidity in the patellar tendon reconstructions makes the BPTB graft less favorable. The results in the 5 years follow-up seem to be equal in the patellar tendon and hamstring tendon autografts in the present study. There is a concern of the appearance of degenerative changes from 2 to 5 years, but on the other hand, we do not know what would have happened without the surgery. Based on the results of this study with a large number of postoperative procedures needed together with obvious knee joint degeneration it can be concluded that ACL injury is not a benign episode and that degenerative arthritis cannot be prevented by ACL reconstruction.

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